

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: Observation Task 1 Part A

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1. TITLE

Design and Develop a College Information Portal Webpage Using HTML and HTML5

2. AIM

To design and develop a static webpage titled “College Information Portal” using HTML and HTML5 that demonstrates the fundamental building blocks of web page creation, including document structure, text formatting, hyperlinks, lists, tables, images, forms, semantic elements and HTML5 input types.

3. OBJECTIVE

The objectives of this experiment are:

- To understand and apply the basic structure of an HTML5 document.
- To display information using headings and paragraphs with appropriate text formatting (Bold, Italic, Superscript, Subscript).
- To create ordered and unordered lists.
- To insert hyperlinks to internal placeholders and external websites.
- To design a table for displaying structured student ranking data.
- To insert and describe an image using the alt attribute.
- To design a registration form using various input controls (text, email, password, date, number, colour, range, radio, checkbox, dropdown, text area).
- To use HTML5 semantic elements (header, nav, section, article, footer) for meaningful page structure.
- To embed audio and video content using HTML5 media elements.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create the structure of web pages. An HTML5 document begins with the `<!DOCTYPE html>` declaration and is organised using nested elements (tags) that define headings, paragraphs, lists, tables, images, links, forms and media.

HTML5 introduced semantic elements such as `<header>`, `<nav>`, `<section>`, `<article>` and `<footer>`, which describe the meaning of the content they contain rather than just its appearance. This improves readability, accessibility and search engine optimisation compared to using generic `<div>` elements everywhere.

Forms are created using the `<form>` element and various `<input>` types. HTML5 added new input types — email, date, number, color and range — which provide built-in validation and richer controls in modern browsers without needing extra JavaScript.

HTML5 also introduced native <audio> and <video> elements, allowing media to be embedded directly in a page with built-in playback controls.

5. ALGORITHM / PROCEDURE

1. Create a new file named index.html.
2. Declare the document type using <!DOCTYPE html> and set up the <html>, <head> and <body> structure.
3. Build a semantic <header> containing a logo <div> and a title <div> with the college name and details.
4. Add a <nav> bar with links to each section of the page and to external sites.
5. Write an “About” <section>/<article> using headings, paragraphs and text-formatting tags (bold, italic, subscript, superscript).
6. Create one unordered list (departments offered) and one ordered list (admission steps).
7. Build a <table> to display the S.No, Name, Branch and CGPA of the top rankers.
8. Design a registration <form> with text, email, password, date, number, colour, range, radio, checkbox, dropdown and textarea inputs, plus submit and reset buttons.
9. Embed an audio file and a video file using the <audio> and <video> elements.
10. Add a semantic <footer> with the copyright line.
11. Save the file and open it in a web browser.
12. Verify that every heading, list, link, table, image, form control and media element renders and behaves as expected.
13. Record the observations against the expected results.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>ANITS College Information Portal</title>

</head>

<body>

  <div class="container">

    <!-- Header -->
    <header>

      <div class="header">

        <!-- Logo -->
        <div class="logo">
          
        </div>

        <!-- College Name and Details -->
        <div class="title">

          <h1>
```

```

        ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY & SCIENCES
    </h1>

    <b>(AUTONOMOUS)</b>

    <p>
        <i>
            Autonomous Institution | Accredited by NBA & NAAC with 'A+' Grade
        </i>
        2X<sub>2</sub>+x<sup>1</sup>
    </p>

    <p>
        Sangivalasa, Bheemunipatnam, Visakhapatnam,
        Andhra Pradesh
    </p>

</div>

</div>

</header>

<!-- Navigation -->
<nav>

    <a href="#">About ANITS</a> |

    <a href="#">Academics</a> |

    <a href="#">Student Data</a> |

    <a href="#">Admission Registration</a> |

    <a href="https://anits.edu.in">
        ANITS Official Website
    </a> |

    <a href="https://www.google.com">
        Google Search
    </a>

</nav>

<hr>

<!-- About Section -->
<section>

    <article>

        <h2>About ANITS</h2>

        <p>

            <b>
                Anil Neerukonda Institute of Technology and Sciences (ANITS)
            </b>

            was established in the academic year 2001-02 by the
            Anil Neerukonda Educational Society (ANES).

            The college is permanently affiliated to
            Andhra University and approved by AICTE.

        </p>

```

Academic Overview

<!-- Departments -->

Offered Departments

Computer Science & Engineering (CSE)

Artificial Intelligence & Machine Learning (AI & ML)

Electronics & Communication Engineering (ECE)

Electrical & Electronics Engineering (EEE)

Information Technology (IT)

Mechanical Engineering

<!-- Admission Steps -->

EAPCET / APICET Admission Steps

Qualify in the AP EAPCET / ICET Entrance Examination.

Register for web counselling and select ANITS
as your college.

Certificate Verification.

Seat Allotment and Fee Payment.

Report to ANITS Campus.

<p>

Visit:

ANITS Official Portal

```
        </a>

    </p>

</article>

</section>

<hr>

<!-- Student Table -->
<section>

    <h2>ANITS Top Rankers</h2>

    <table>

        <tr>

            <th>S.No</th>

            <th>Name</th>

            <th>Branch</th>

            <th>CGPA</th>

        </tr>

        <tr>

            <td>1</td>

            <td>Aruna Devi</td>

            <td>AI & ML</td>

            <td>9.82</td>

        </tr>

        <tr>

            <td>2</td>

            <td>Sai Kumar</td>

            <td>CSE</td>

            <td>9.76</td>

        </tr>

        <tr>

            <td>3</td>

            <td>Harika</td>

            <td>ECE</td>

            <td>9.69</td>

        </tr>

    </table>

</section>
```

```
<tr>
    <td>4</td>
    <td>Rahul</td>
    <td>IT</td>
    <td>9.60</td>
</tr>
</table>
</section>
<hr>

<!-- Student Registration Form -->
<section>
    <h2>Student Registration Form</h2>
    <form>
        <label>Name:</label>
        <br>
        <input type="text" required>
        <br><br>

        <label>Email:</label>
        <br>
        <input type="email" required>
        <br><br>

        <label>Password:</label>
        <br>
        <input type="password" required>
        <br><br>

        <label>Date of Birth:</label>
        <br>
        <input type="date">
        <br><br>

        <label>Age:</label>
        <br>
        <input type="number" min="16" max="30">
```

```
<br><br>

<label>Favorite Color:</label>

<br>

<input type="color">

<br><br>

<label>Programming Skill Level:</label>

<br>

<input type="range" min="0" max="100">

<br><br>

<label>Gender:</label>

<br>

<input type="radio" name="gender"> Male
<input type="radio" name="gender"> Female
<input type="radio" name="gender"> Other

<br><br>

<label>Hobbies:</label>

<br>

<input type="checkbox"> Reading
<input type="checkbox"> Coding
<input type="checkbox"> Sports
<input type="checkbox"> Music

<br><br>

<label>Branch:</label>

<br>

<select>
  <option>AI & ML</option>
  <option>CSE</option>
  <option>ECE</option>
  <option>EEE</option>
  <option>IT</option>
</select>

<br><br>
```

```
<label>Comments:</label>

<br>

<textarea rows="5" cols="45"></textarea>

<br><br>

<input type="submit" value="Submit">

<input type="reset" value="Reset">

</form>

</section>

<hr>

<!-- HTML5 Multimedia -->
<section>

  <h2>College Anthem (Audio)</h2>

  <audio controls>

    <source src="college.mp3" type="audio/mpeg">

    Your browser does not support audio.

  </audio>

  <br><br>

  <h2>College Campus Video</h2>

  <video width="500" controls>

    <source src="campus.mp4" type="video/mp4">

    Your browser does not support video.

  </video>

</section>

<hr>

<!-- Footer -->
<footer>

  <p>
    © 2026 ANITS - College Information Portal
  </p>

</footer>

</div>

</body>

</html>
```


7. CODE EXPLANATION

Tag / Attribute	Description
<!DOCTYPE html>	Declares the document as an HTML5 document.
<header>, <nav>, <section>, <article>, <footer>	HTML5 semantic elements used to structure the page into meaningful regions (logo/title block, navigation, About/Academics article, table section, form section, media section, footer).
<div class="...">	Generic containers used inside the header to group the logo and the title/details block.
, <i>	Apply bold and italic text formatting (college name, "(AUTONOMOUS)", accreditation line).
<sub>, <sup>	Render subscript and superscript characters, demonstrated with the sample expression $2X_2 + x^1$.
<h1>, <h2>, <h3>	Define the main heading, section headings (About ANITS, Academic Overview, Top Rankers) and sub-headings (Offered Departments, Admission Steps).
,	Creates the unordered list of offered departments.
,	Creates the ordered list of EAPCET / APICET admission steps.
	Creates hyperlinks to internal placeholders and external sites (ANITS Official Website, Google Search).
<table>, <tr>, <th>, <td>	Builds the "ANITS Top Rankers" table showing S.No, Name, Branch and CGPA.
	Embeds the college logo with descriptive alternate text.
<form>, <label>, <input>, <select>, <textarea>	Collect user input through various form controls in the Student Registration Form.
type="radio" / "checkbox"	Allow single-choice (Gender) and multiple-choice (Hobbies) selection.
type="email", "password", "date", "number", "color", "range"	HTML5 input types that provide built-in validation and richer controls (email format check, masked password, date picker, numeric min/max, colour swatch, slider).
<select>, <option>	Dropdown list used to choose the Branch.
<audio controls>, <video controls>	Embed playable audio (college anthem) and video (campus video) content with native browser controls.
<hr>	Draws a horizontal divider between page sections.

8. TEST CASES AND EXECUTION DATA

The page was opened in a browser and each feature was individually verified:

Test Case	Feature Tested	Action Performed	Expected Result	Status
TC-01	Page structure	Open the page in a browser	Header, nav, all sections and footer render correctly	Pass
TC-02	Text formatting	View the About section	Bold, italic, subscript and superscript text display correctly	Pass
TC-03	Lists	View the Academic Overview section	Unordered department list and ordered admission-steps list display	Pass

Test Case	Feature Tested	Action Performed	Expected Result	Status
			correctly	
TC-04	Hyperlinks	Click ANITS Official Website and Google Search links	Each link opens the corresponding site	Pass
TC-05	Table	View the ANITS Top Rankers table	S.No, Name, Branch and CGPA display in a structured table	Pass
TC-06	Image	View the college logo	Logo renders with correct alt text on hover / load failure	Pass
TC-07	Form validation	Submit the form with Name/Email left empty	Browser blocks submission and highlights required fields	Pass
TC-08	Radio buttons	Select Male / Female / Other	Only one gender option can be selected at a time	Pass
TC-09	Checkboxes	Select multiple hobbies	Multiple hobby options can be selected simultaneously	Pass
TC-10	Dropdown	Select a branch from the list	Selected branch value updates correctly	Pass
TC-11	Color & Range inputs	Pick a favourite colour and drag the skill-level slider	Colour swatch and slider value update live	Pass
TC-12	Date & Number inputs	Pick a date of birth and enter an age	Date picker opens; number field respects the 16-30 range	Pass
TC-13	Audio / Video	Play the embedded audio and video	Native browser controls appear and media plays (sample media files)	Pass

9. OUTPUT

Actual output produced by running the program above in a browser:



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(AUTONOMOUS)

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Sangivalasa, Bheemunipatnam, Visakhapatnam, Andhra Pradesh

[About ANITS](#) | [Academics](#) | [Student Data](#) | [Admission Registration](#) | [ANITS Official Website](#) | [Google Search](#)

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ANITS Top Rankers

S.No	Name	Branch	CGPA
1	Aruna Devi	AI & ML	9.82
2	Sai Kumar	CSE	9.76
3	Harika	ECE	9.69
4	Rahul	IT	9.60

Student Registration Form

Name:

Email:

Password:

Date of Birth:

Age:

Favorite Color:

Programming Skill Level:

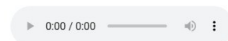
Gender:
☐ Male ☐ Female ☐ Other

Hobbies:
☐ Reading ☐ Coding ☐ Sports ☐ Music

Branch:

Comments:

College Anthem (Audio)



College Campus Video



10. RESULT

Thus, a webpage titled “College Information Portal” was successfully designed and developed using HTML and HTML5. Basic HTML tags, text formatting, ordered and unordered lists, hyperlinks, tables, an image, a registration form with multiple input types, HTML5 semantic elements, and audio/video embedding were implemented and verified through browser testing, matching the actual output shown above.